AP CSP Unit 6 Day 8 Precision and Limits Assessment

1. (DOK 1) What is meant by precision in computing?

A. The speed of a computer

B. The level of exactness in a value or calculation

C. The amount of storage available

D. The number of programs running

2. (DOK 1) Which type of number can contain a decimal portion?

A. Integer

B. Floating-point

C. Binary digit

D. Hex digit

3. (DOK 1) What does overflow occur when?

A. A value exceeds the storage capacity available

B. A file is deleted

C. A program starts

D. A variable is named incorrectly

4. (DOK 1) Which notation is represented by 7.5e2?

A. Binary notation

B. Hexadecimal notation

C. Scientific notation

D. Unicode notation

5. (DOK 2) Why might 0.1 + 0.2 not equal exactly 0.3 in some programs?

A. The computer is broken

B. Floating-point values are approximations

C. Integers are too large

D. The program uses Unicode

6. (DOK 2) What is the value of 7.5e2?

A. 75

B. 750

C. 7500

D. 0.75

7. (DOK 2) Why are floating-point numbers useful despite approximation errors?

A. They represent a wide range of values efficiently

B. They eliminate all errors

C. They require no memory

D. They only store whole numbers

8. (DOK 3) A game score suddenly changes to a negative number after exceeding a system limit. What is the most likely cause?

A. Unicode conversion

B. Overflow

C. Rounding to nearest tenth

D. Data compression

9. (DOK 3) A programmer notices small differences between expected and actual results in repeated calculations. What is the best explanation?

A. Network issues

B. Floating-point precision limitations

C. A syntax error

D. ASCII encoding

10. (DOK 4) A MakeCode Arcade project tracks distance traveled using many decimal calculations. Which approach best demonstrates understanding of precision limitations?

A. Assume every calculation is exact

B. Test for rounding effects, evaluate acceptable error ranges, and choose data types carefully

C. Convert all values to text

D. Store every value as a character